Real Analysis

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Lecture 1: Introduction and Proofs

Logic

Let's say we have proposition P and Q. They can be joined by "and", "or", "implies", or "if and only if".

For example, if P is a proposition, then "not P" is a new propostion that is true whenever P is false and vice versa. The symbolic representation for "not P" is $\neg P$ or $\overline{P}$

Two propositions, P and Q, can be joined by "and", "or", "implies", or "if and only if" to form a new proposition. The truth of this new proposition is determined by the truth of P and Q according to the **Truth Table**.

P	Q	"P and Q" $(P \wedge Q)$	"P or Q" $(P \Rightarrow Q)$	"P implies Q or" "if P, then Q" $(P \Rightarrow Q)$	"P if and only if Q" or "P iff Q" $(P \Leftrightarrow Q)$
F	F	F	F	T	T
F	T	F	T	T	F
T	F	F	T	F	F
T	T	T	T	T	T

Here are a few hidden features within this table:

- The phrase "P or Q" is true if P is true, Q, or both are true.
- The phrase "P implies Q" is true when P is false or Q is true.

There are two more important phrases in mathematical writing: "for all" (symbolized by $\forall$) and "there exists" (symbolized by $\exists$). These are called ***quantifiers***.

Definition 1 (Quantifiers)**.** A quantifier is always followed by a variable (and perhaps an indication of the range of that variable) and then a predicate, which typically involves that variable. Here are a couple of examples:

$$\forall x \in \mathbb{R}^+ \quad e^x < (1+x)^{1+x} \tag{1}$$
$$\exists n \in \mathbb{N} \quad 2^n > (100n)^{100}. \tag{2}$$

The first statement says that e^x is less than $(1+x)^{1+x}$ for every positive real number x. The second statement says that there exists a natural number n such that $2^n > (100n)^{100}$.

The special symbols such as $\forall, \exists, \neg$, and $\wedge$ are useful to logicians who are trying to express mathematicians ideas without resorting to the En-

glish language at all. Also, other mathematicians use these symbols as shorthands.

Proving an Implication

Let's try to prove the following theorem:

Theorem 1 (Let $P(a,b)$ be any predicate defined for all $a \in \mathbb{A}$ and $b \in (B)$ Then:)**.**

$$(\exists a \in \mathbb{A} \quad \forall b \in \mathbb{B} \quad P(a,b)) \quad \Rightarrow \quad (\forall b \in \mathbb{B} \quad \exists a \in \mathbb{A} \quad P(a,b)). \tag{3}$$

Let's impose a specific interpretation in order to give concrete meaning:

$$\begin{aligned} \mathbb{A} &= \{6.042 \quad \textit{students}\} \\ \mathbb{B} &= \{6.042 \quad \textit{lectures}\}. \\ P(a,b) &= \text{"student } a \textit{ falls asleep during lecture } b\text{"} \end{aligned} \tag{4}$$

Interpreting the left side in these terms give:

$$\begin{aligned} &\exists a \in \mathbb{A} \quad \forall b \in \mathbb{B} \quad P(a,b) \\ &= \text{"there exists a student that falls asleep in every lecture"}. \end{aligned} \tag{5}$$

So, this side states that some particular student always falls asleep. Let's call him Snoozer. Now, here's the right side:

$$\begin{aligned} &\forall b \in \mathbb{B} \quad \exists a \in \mathbb{A} \quad P(a,b) \\ &= \text{"in every lecture, some student falls asleep"}. \end{aligned} \tag{6}$$

This is slightly different than the left side because there might be a different sleeper in each lecture. The left side should imply the right. If Snoozer sleeps in every lecture, then in every lecture some student is surely asleep.